
NHS BENCHMARK COMPARISON REPORT

(ANA-4)

Comparing CSEW Self-Reported Mental Health Rates with NHS Referral Trends (2019–2024)

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Project: AI Mental Health Assistant

COMPARING CSEW SELF-REPORTED MENTAL HEALTH RATES WITH NHS REFERRAL TRENDS (2019–2024)

1. OBJECTIVE

The objective of this analysis was to compare self-reported mental health prevalence trends from the Crime Survey for England and Wales (CSEW) with NHS mental health referral trends between 2019 and 2024.

The study focuses on identifying:

- differences between reported mental health burden and healthcare engagement,
- periods of significant increase or decline,
- gaps between population distress levels and NHS referral activity,
- alignment between national mental health trends and healthcare system response.

2. DATASETS USED:

2.1 CSEW Mental Health Dataset

Source: Office for National Statistics (ONS)

Dataset contains annual self-reported mental health indicators including:

- Anxiety
- Depression
- Low Wellbeing

Time Period:
2019–2024

Variables used:

Variable	Description
Year	Reporting year
Anxiety	Percentage reporting high anxiety
Depression	Percentage reporting depression symptoms
Low Wellbeing	Percentage reporting reduced wellbeing

2.2 NHS Referral Benchmark Dataset

Source: NHS England Mental Health Statistics

Dataset contains estimated annual mental health referral trend percentages used as benchmark indicators for healthcare system engagement.

Variables used:

Variable	Description
Year	Reporting year
NHS_Referral_Rate	Estimated annual mental health referral percentage

3. TOOLS & TECHNOLOGIES USED:

Tool	Purpose
Python	Data analysis
Pandas	Data cleaning and merging
Matplotlib	Trend visualisation
Seaborn	Statistical plotting
Jupyter Notebook	Workflow execution
GitHub	Version control and project submission

4. METHODOLOGY:

The analysis was completed in four stages:

1. Imported both datasets into Python using Pandas.
2. Merged datasets using the shared Year variable.
3. Compared annual CSEW mental health prevalence rates with NHS referral benchmarks.
4. Calculated referral gaps using comparative percentage analysis.

The following formula was used:

Gap = Anxiety Rate–NHS Referral Rate

This gap represents the difference between self-reported mental health burden and healthcare engagement levels.

5. GAP-ANALYSIS TABLE:

Year	Anxiety (%)	Depression (%)	NHS Referral Rate (%)	Gap (%)
2019	12.1	10.5	9.8	2.3
2020	18.4	15.2	11.5	6.9
2021	21.2	18.9	15.2	6.0
2022	19.5	17.1	14.1	5.4
2023	17.8	15.6	13.4	4.4
2024	16.9	14.8	12.9	4.0

6. FINDINGS AND INTERPRETATION:

6.1 COVID-Era Increase in Mental Health Burden

A sharp rise in anxiety and depression was observed during 2020 and 2021.

- Anxiety increased from 12.1% in 2019 to 21.2% in 2021.
- Depression increased from 10.5% to 18.9% during the same period.

This trend aligns with the social and economic impact of the COVID-19 pandemic, including:

- lockdown restrictions,
- financial instability,
- social isolation,
- uncertainty regarding health and employment.

6.2 NHS Referrals Increased More Slowly

Although NHS referral rates increased after 2019, the increase was smaller than the rise observed in self-reported distress.

For example:

- Anxiety increased by approximately 9 percentage points between 2019 and 2021.
- NHS referrals increased by around 5 percentage points during the same period.

This indicates that a substantial proportion of individuals experiencing mental health symptoms may not have accessed formal mental health services.

6.3 Largest Referral Gap During Pandemic Period

The highest gap was recorded in 2020:

Year Gap (%)

2020 6.9

This suggests that mental health distress increased faster than healthcare utilisation during the pandemic period.

Possible contributing factors include:

- reduced healthcare accessibility,
- waiting times,
- delayed treatment seeking,
- limited awareness of support services,
- social stigma associated with mental health treatment.

6.4 Gradual Stabilisation After 2022

From 2022 onward:

- anxiety and depression levels gradually declined,
- NHS referral rates remained relatively stable,
- overall gap values reduced slightly.

However, the gap did not disappear entirely, indicating that unmet mental health needs may still persist within the population.

7. VISUALIZATION SUMMARY:

The generated comparison charts showed:

- a clear rise in mental health indicators during the pandemic period,
- NHS referral trends following a similar direction,
- persistent separation between reported distress and referral activity.

The line comparison chart effectively highlighted differences in trend magnitude between population-level symptoms and healthcare engagement.

8. LIMITATIONS:

Several limitations should be considered:

1. CSEW measures self-reported symptoms, not clinical diagnoses.
2. NHS referral percentages represent healthcare engagement rather than total prevalence.
3. Differences in survey methodology and reporting criteria may affect direct comparability.
4. Referral data used in this analysis serves as a benchmark approximation for analytical comparison.

9. CONCLUSION

The analysis identified a consistent gap between self-reported mental health burden and NHS referral activity between 2019 and 2024.

Mental health indicators increased sharply during the COVID-19 period, while NHS referrals rose at a slower rate. Although trends improved after 2022, the gap between distress levels and healthcare engagement remained visible.

The findings suggest that periods of increased population distress may place pressure on healthcare systems and may also highlight barriers to accessing mental health support services.